Lecture 16

2/25/26
JRA

Housekeeping

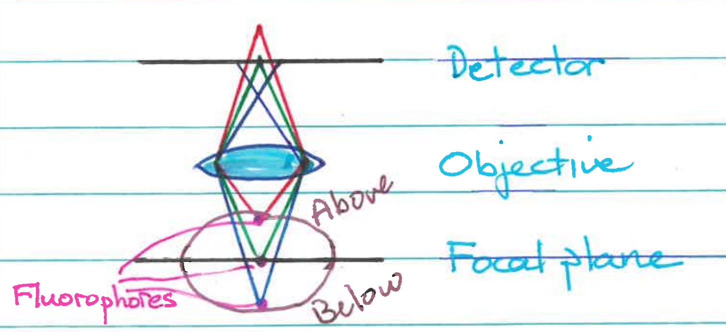
- Midterm will be take home (in ~2 wks)
- Next homework due Sunday at midnight
- Need Fiji / ImageJ for PSF homework
- We'll look briefly at polarization on Friday

Blur Reduction & 3D Imaging

Look today at blur problem, blur reduction, & 3D imaging

Goals:
- Image specimen (e.g., cell)
- Address blur problem
- Do 3D imaging (take a series of optical sections, usually bottom to top, to generate a "z stack")

Blur Problem



- Fluorophores distributed at various depths in sample
- One plane (the "focal plane") where light in focus
- Collect light from above & below ⇒ blur spots

Light from all three fluorophores in drawing will contribute to image

out of focus

RED (from above focal plane) - OOF (blurred)
GREEN (from focal plane) - in focus
BLUE (from below focal plane) - OOF (blurred)

Here, the light (GREEN) that is in focus will be heavily contaminated with light (RED & BLUE) that is out of focus.

The resulting image will be blurred.

Blurring is NOT a problem if the sample is very thin, because all fluorophores will be confined to the same plane. However, even cells (~5-10 μm thick) are thick enough so their fluorescence images are plagued by blur

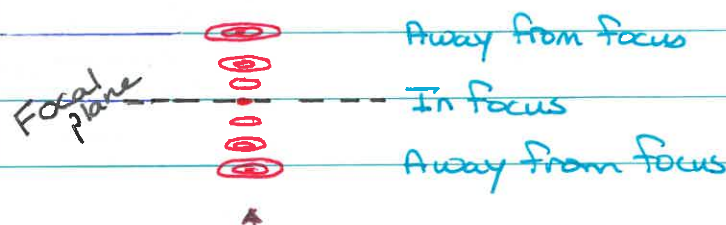
Point Spread Function (PSF)

Another problem arises due to how a fluorescence microscope images a point source of light (see HW A6.1).

Ideally, the image of a point source of light would be a point (in focus or visible in just one plane)

Unfortunately, this is <u>NOT TRUE</u>.

Look at image of point source when it is in & out of focus



See light from point source even when it's way out of focus

circular diffraction pattern above & below focus

The 3D image of a point source of light is called a Point Spread Function (PSF) because the point is spreading over space.

Need to eliminate the 3D spreading to get clean (reduced-blur) images.

<u>In Focus</u> – get a tiny bright dot image ~0.2 μm radius (w/ blue light)

<u>Out of Focus</u> – get a smeared-out dimmer image ringing like a circular diffraction pattern BLUR SPOT

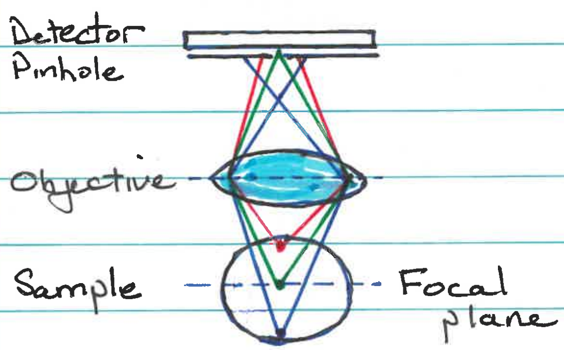
The total light intensity is the same, whether in or out of focus, because it's the same total amount of light traveling thru focus

HW A6.1 - do homework activity in ImageJ, adding up total light in each section of a measured PSF to confirm it's constant

Laser Scanning Confocal Microscopy (LSCM)

A first solution to the blur problem is laser scanning confocal microscopy

The idea of confocal microscopy was introduced in the 1950s by Marvin Minsky.



- Red light, originating above focus, is mostly BLOCKED
- Green light, originating at focus, gets thru pinhole & is DETECTED
- Blue light, originating above focus, is mostly BLOCKED

Key idea: Put a pinhole in an intermediate image plane, between the sample & the detector, that preferentially blocks the out-of-focus light. Here, light (GREEN) from the focal plane passes thru the pinhole, where the light is again in focus, & light (RED, BLUE) from other planes is (mostly) blocked

We discussed methods a few lectures ago that involve filtering in the BFP of the objective. Examples are darkfield (DF) & phase contrast (PC) microscopies. There the image was completly out of focus when filtered.

Confocal μscopy is another filtering method, which is performed in an image plane (i.e., in a plane where the image is in focus).

Implementation requires an ability to illuminate a tiny spot.

Technical Limitations

1) Illumination

Image extended specimens point-by-point by using objective to focus a laser beam to a diffraction-limited spot. The size of the spot depends on the color of the light (BLUE = small spot, RED = large spot).

Typical spot size $\approx 0.5\,\mu m$ diameter

2) Detection

Fluorophores in the illuminated spot (& light path in general) will fluoresce.

The objective collects fluorescence & focuses it to a spot in an intermediate image plane.

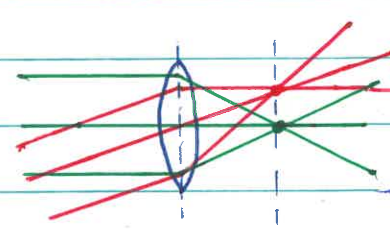
The pinhole is positioned in the intermediate image plane, where it passes the focused light & mostly blocks the remaining (out-of-focus) light coming from above & below the confocal "spot"

3) Raster Scanning

Steps 1 & 2 (above) only involve a single spot in the sample.

An image is created by scanning the spot over the extended specimen.

Scanning is performed using pivotable mirrors in the light path that alter the angle with which light hits the objective, which in turn alters the position of the spot



Angle of incident light determines position of spot in focal plane

The scanning mirrors don't move during the time it takes to generate & collect fluorescence (because it happens so quickly). The fluorescence follows the same path thru the mirrors taken by the excitation light, only backwards, so stationary pinhole can be used to reject out-of-focus light for all spot positions.

LSCM is very powerful but also has limitations

LSCM - Strengths

Deblurring • Generates highly deblurred images

Simplicity • Very simple for user to implement (because it's commercially available)

3D Imaging • Can generate 3D images by moving the stage (& so the specimen) so different planes come into view

LSCM - Limitations

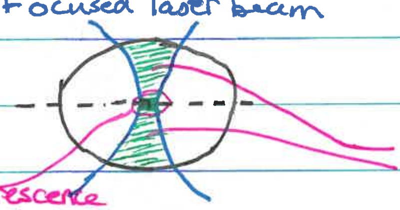
Slow • Slow because you need to scan
Takes about 1 second to collect a single 2D image)
Too slow for fast biological processes

Photobleaching
Phototoxicity

- Illuminates all depths in specimen but collects light only from in-focus plane (due to pinhole). This means there can be photobleaching &/or phototoxicity to parts of the sample that are NOT contributing to the signal.

Focused laser beam

Focused beam has an hourglass shape



Collect fluorescence only from beam waist

Illuminate all depths { above / below
→ photobleaching
→ phototoxicity
These depths do NOT contribute to signal

$\boxed{\text{VIDEO}}$ Kurt Thorn Confocal μscopy

Viewed selected portions of video, especially relating to scanning mechanism & limitations

Scanning {
- Pinhole stationary
- Mirrors scan excitation
- Mirrors "descan" emission

Showed geometry for real μscope setup, including light sources, scan head, etc.

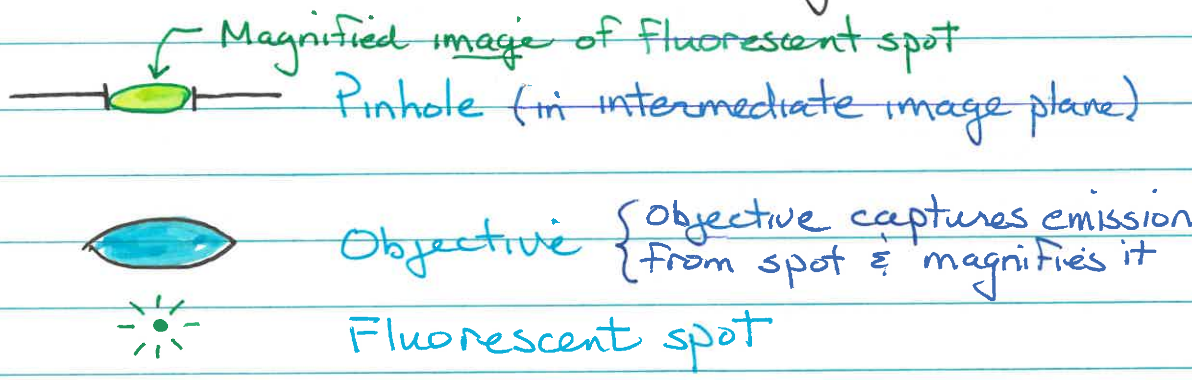
Limitations {
- Slow – scan point-by-point
- Low light efficiency {
  - Uses PMT (25% efficient)
  - vs camera (90% efficient)
  - Bad for dim samples

Pinhole Size

(See HW Problem 6.15)

How large should the pinhole be?

The smaller the pinhole, the larger the amount of out-of-focus light that will be rejected. However, too small, and the signal will vanish!



The pinhole should match the magnified spot size in the intermediate image plane. This will pass the bulk of light from the fluorescent spot.

Example | $0.5\ \mu m$ spot
$100\times$ objective

Pinhole diameter $= 100 \cdot 0.5\mu m = \underline{50\mu m}$